

[문제풀이] Deployment 업데이트 및 롤백 실행하기-1

- 문제

kubernetes.io		
Keyword	deployment	kubectl rollout

TASK

Please complete the following:

- Configure Deployment **app** in the Namespace **rollout** with a **maxSurge of 75%** and a **maxUnavailable of 50%**
- Update the app Deployment to use version **tag 2.0** and name **app-v2** for the container.
- Perform a rollback of the app deployment to the previous version

- 배포되어있는 deployment를 아래와 같이 업데이트
 - maxSurge: 75%
 - maxUnavailable: 50%
 - pod의 이미지 버전 tag를 2.0으로 변경
 - pod의 container 이름을 app-v2 로 변경
- 업데이트 후, 해당 deployment를 다시 이전 버전으로 rollback

- 정답

- `kubectl edit deploy` 명령어로, Deployment를 아래와 같이 update

```
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "1"
  creationTimestamp: "2026-07-13T18:13:03Z"
```

```

generation: 1
name: app
namespace: rollout
resourceVersion: "25475"
uid: 28cba55d-77d6-40ea-ae87-ba64d94fef9
spec:
  progressDeadlineSeconds: 600
  replicas: 4
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      name: app
  strategy:
    rollingUpdate:
      maxSurge: 75% # 변경
      maxUnavailable: 50% # 변경
    type: RollingUpdate
  template:
    metadata:
      labels:
        name: app
    spec:
      containers:
      - image: lpro/exam:2.0 # 변경
        imagePullPolicy: IfNotPresent
        name: app-v2 # 변경
        ports:
        - containerPort: 8080
          protocol: TCP
        resources: {}
        terminationMessagePath: /dev/termination-log
        terminationMessagePolicy: File
      dnsPolicy: ClusterFirst
      restartPolicy: Always
      schedulerName: default-scheduler
      securityContext: {}
      terminationGracePeriodSeconds: 30

```

■ Deployment update하기

- maxSurge

- 업데이트 중 원래 replicas보다 추가로 생성할 수 있는 Pod의 최대 개수
- 75%로 지정하면, replicas가 4이므로, 업데이트 중 최대 3개 pod가 추가 생성될 수 있음

- maxUnavailable

- 업데이트 중 사용 불가능(Unavailable)해도 되는 Pod의 최대 개수
- 50%로 지정하면, replicas가 4이므로, 최대 2개 pod까지만 Unavailable 가능
 - 즉, update 과정에서 최소한 원래 개수 4 - maxUnavailable개수 2 = 2개 pod는 ready 상태여야 함

■ Rollback하기

```
kubectl rollout undo deploy -n rollout
```

- deployment 문서에, deployment를 이전 버전으로 rollback하는 가이드가 있음
<https://kubernetes.io/docs/concepts/workloads/controllers/deployment/>

The screenshot shows the 'Deployments' page in the Kubernetes documentation. The left sidebar contains a navigation menu with categories like Documentation, Getting started, Concepts, and Workload Management. Under Workload Management, 'Deployments' is highlighted. The main content area has a heading 'Deployments' followed by a description: 'A Deployment manages a set of Pods to run an application workload, usually one that doesn't maintain state.' It also mentions that a Deployment provides declarative updates for Pods and ReplicaSets. A 'Note' box states: 'Do not manage ReplicaSets owned by a Deployment. Consider opening an issue in the main Kubernetes repository if your use case is not covered below.' Below this is a 'Use Case' section with the text: 'The following are typical use cases for Deployments:' followed by a list of four use cases: 1. Create a Deployment to rollout a ReplicaSet. 2. Declare the new state of the Pods by updating the PodTemplateSpec of the Deployment. 3. Rollback to an earlier Deployment revision if the current state of the Deployment is not stable. 4. Scale up the Deployment to facilitate more load.

- 위 링크를 누르면 rollback하는 명령어가 바로 있지는 않고, 예시 case에 대한 설명과 rollout history를 보는 방식을 먼저 설명하고, 그 다음에 rollback하는 명령어가 나온다.

The screenshot shows the 'Rolling Back to a Previous Revision' section in the Kubernetes documentation. The left sidebar is the same as the previous screenshot, with 'Deployments' highlighted. The main content area has a heading 'Rolling Back to a Previous Revision' followed by the text: 'Follow the steps given below to rollback the Deployment from the current version to the previous version, which is version 2.' A numbered list starts with '1. Now you've decided to undo the current rollout and rollback to the previous revision:'. Below this is a code block showing the command: `kubectl rollout undo deployment/nginx-deployment`. The text 'The output is similar to this:' is followed by a code block showing the output: `deployment.apps/nginx-deployment rolled back`. Below this, the text 'Alternatively, you can rollback to a specific revision by specifying it with --to-revision:' is followed by a code block showing the command: `kubectl rollout undo deployment/nginx-deployment --to-revision=2`.

- kubectl rollout 명령어
https://kubernetes.io/docs/reference/kubectl/generated/kubectl_rollout/

- deployment, demonset, statefulset 등, 배포에 관련된 리소스들의 배포 상태를 관리하는 명령어

kubectl rollout

Manage the rollout of a resource

Synopsis

Manage the rollout of one or many resources.

Valid resource types include:

- deployments
- daemonsets
- statefulsets

```
kubectl rollout SUBCOMMAND
```

- See Also에, 가능한 모든 kubectl rollout xxxx 명령어 목록이 있다.

See Also

- [kubectl](#) - kubectl controls the Kubernetes cluster manager
 - [kubectl rollout history](#) - View rollout history
 - [kubectl rollout pause](#) - Mark the provided resource as paused
 - [kubectl rollout restart](#) - Restart a resource
 - [kubectl rollout resume](#) - Resume a paused resource
 - [kubectl rollout status](#) - Show the status of the rollout
 - [kubectl rollout undo](#) - Undo a previous rollout
- 리소스 타입과 이름을 모두 지정하거나, 리소스 타입만 지정할 수 있음

kubectl rollout undo

Undo a previous rollout

Synopsis

Roll back to a previous rollout.

```
kubectl rollout undo (TYPE NAME | TYPE/NAME) [flags]
```

Examples

```
# Roll back to the previous deployment
kubectl rollout undo deployment/abc

# Roll back to daemonset revision 3
kubectl rollout undo daemonset/abc --to-revision=3

# Roll back to the previous deployment with dry-run
kubectl rollout undo --dry-run=server deployment/abc
```

◦ 리소스 타입만 지정

```
[candidate@k8s-master ~]$ kubectl rollout status deployment -n rollout
deployment "app" successfully rolled out
[candidate@k8s-master ~]$ kubectl rollout history deployment -n rollout
deployment.apps/app
REVISION  CHANGE-CAUSE
1          <none>
2          <none>
```

```
[candidate@k8s-master ~]$ kubectl rollout undo deployment -n rollout
deployment.apps/app rolled back
[candidate@k8s-master ~]$ kubectl rollout status deployment -n rollout
deployment "app" successfully rolled out
[candidate@k8s-master ~]$ kubectl rollout history deployment -n rollout
deployment.apps/app
REVISION  CHANGE-CAUSE
2          <none>
3          <none>
```

◦ 리소스 타입과 이름을 모두 지정

```
[candidate@k8s-master ~]$ k rollout status -n rollout deployment/app
deployment "app" successfully rolled out
[candidate@k8s-master ~]$ k rollout history -n rollout deployment/app
deployment.apps/app
REVISION  CHANGE-CAUSE
1          <none>
2          <none>
```

```
[candidate@k8s-master ~]$ k rollout undo -n rollout deployment/app
deployment.apps/app rolled back
[candidate@k8s-master ~]$ k rollout history -n rollout deployment/app
deployment.apps/app
REVISION  CHANGE-CAUSE
2          <none>
3          <none>
```

- 해당 namespace에 deployment가 여러 개 있는 경우, 리소스 타입만 지정하면, 아래와 같이 'rollout' namespace에 있는 모든 deployment가 조회되고, rollback 된다.

```
[candidate@k8s-master ~]$ k get deploy -n rollout
NAME          READY    UP-TO-DATE    AVAILABLE    AGE
app            1/1      1              1            6m15s
app999         1/1      1              1            4m10s
```

```
[candidate@k8s-master ~]$ k rollout history -n rollout deployment
deployment.apps/app
REVISION  CHANGE-CAUSE
1          <none>
2          <none>

deployment.apps/app999
REVISION  CHANGE-CAUSE
1          <none>
2          <none>
```